

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL A-LEVEL PHYSICS

Unit 3 Fields and their consequences

Wednesday 15 January 2025

07:00 GMT

Time allowed: 2 hours

Materials

For this paper you must have:

- a Data and Formulae Booklet as a loose insert
- a ruler with millimetre measurements
- a scientific calculator, which you are expected to use where appropriate
- a protractor.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each question or on blank pages.
- All working must be shown.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 80.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8–22	
TOTAL	

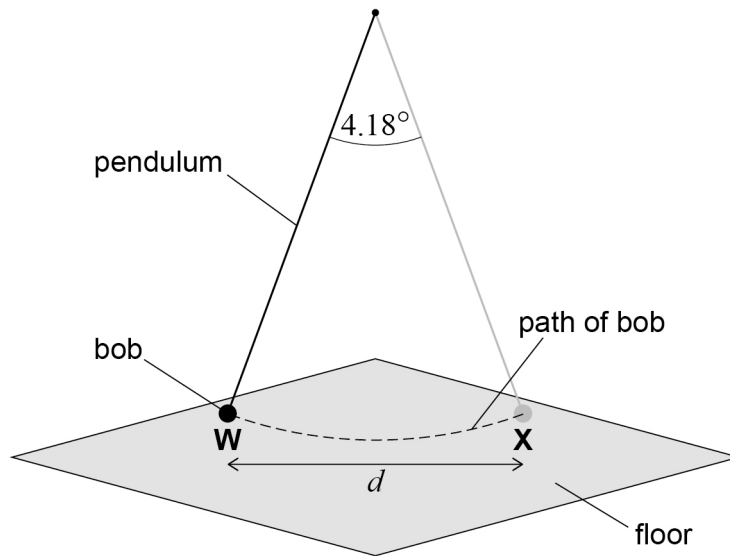


Section AAnswer **all** questions in this section.**0 1**

Figure 1 shows a simple pendulum of length 67.7 m .
The pendulum bob is released from position **W** and swings in a vertical plane.

Figure 1

not to scale



The pendulum swings through an angle of 4.18° . The bob just reaches position **X**.
The horizontal distance between **W** and **X** is d .

Air resistance is negligible.

0 1 . 1Calculate d .**[2 marks]**
 $d =$ _____ m


0 1 . 2 Calculate the maximum speed of the bob.

[3 marks]

maximum speed = _____ m s^{-1}

Question 1 continues on the next page

Turn over ►

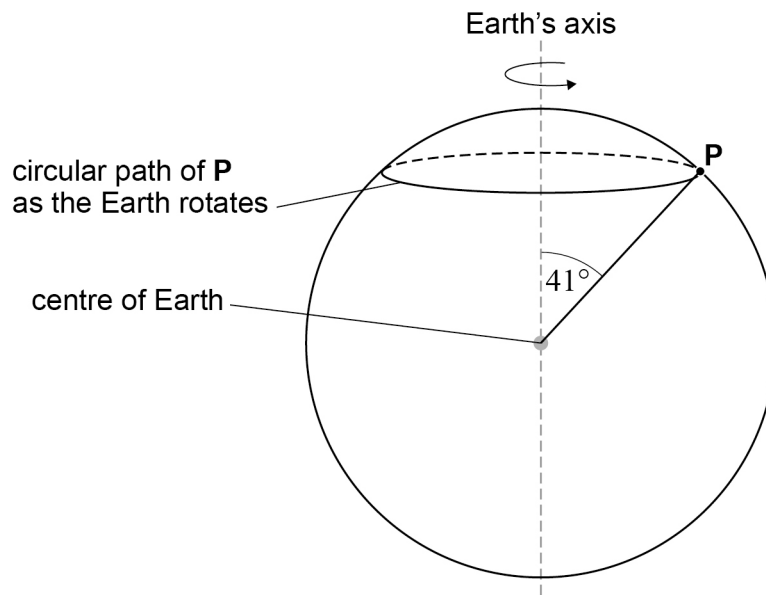


The pendulum is at point **P** on the surface of the Earth.

0 1 . 3 Figure 2 shows **P**.

The line joining **P** to the centre of the Earth is at an angle of 41° to the Earth's axis.
Due to the Earth's rotation, **P** travels in a circular path about the Earth's axis.

Figure 2



Calculate the linear speed of **P** due to the rotation of the Earth.

[2 marks]

speed = _____ m s^{-1}



0 1 . 4

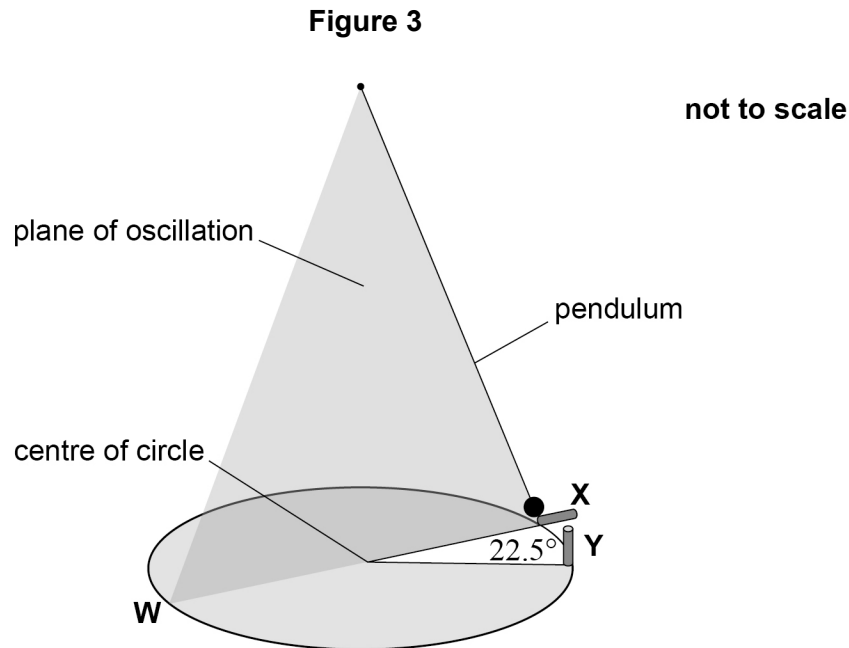
Due to the rotation of the Earth, the plane of oscillation of the pendulum rotates slowly about a vertical axis relative to the floor.

The plane of oscillation rotates through 360° in 31.8 hours.

Pegs are placed on the floor at positions **X** and **Y** on a circle.

The angle between **X** and **Y**, measured at the centre of the circle, is 22.5° as shown in **Figure 3**.

Figure 3 shows the pendulum just after it knocks over the peg at **X**.



The plane of oscillation continues to rotate relative to the floor. The amplitude of oscillation of the pendulum is unchanged.

After knocking over the peg at **X**, the pendulum goes through n oscillations before knocking over the peg at **Y**.

Calculate n .

[3 marks]

$n =$ _____

10

Turn over ►



0 2

An unpowered spacecraft **B** orbits the Sun.

The orbit is not circular.

Assume that the only gravitational force on **B** is due to the Sun.

The following data are about **B**:

mass of **B** = 1600 kg

smallest distance between **B** and the centre of the Sun = 4.20×10^{10} m

largest distance between **B** and the centre of the Sun = 1.37×10^{11} m.

0 2 . 1

Calculate the maximum gravitational force of the Sun on **B**.

[2 marks]

maximum gravitational force = _____ N

0 2 . 2

The total orbital energy of a satellite is the sum of its kinetic energy and its gravitational potential energy.

The total energy of **B** is -1.19×10^{12} J.

Calculate the minimum speed of **B**.

[3 marks]

minimum speed = _____ m s^{-1}



An unpowered spacecraft **C** is in a circular orbit of the Sun.
C has an orbital period of 1.45×10^7 s.

0 2 . 3

Calculate the orbital radius of **C**.

[3 marks]

orbital radius = _____ m

0 2 . 4

Explain which spacecraft, **B** or **C**, is moving on an equipotential.

[1 mark]

9

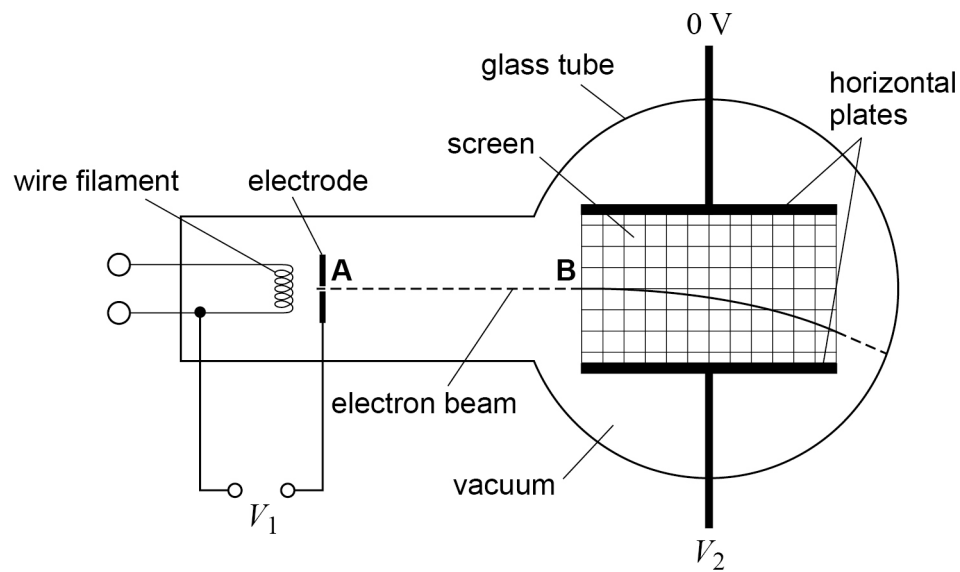
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0 3

Figure 4 shows an arrangement that is used to investigate the properties of electrons.

Figure 4

Electrons are emitted from a wire filament with a negligible initial speed.
The electrons are accelerated through a potential difference V_1 towards an electrode.
The electrons that pass through the hole at **A** have a speed of $1.48 \times 10^7 \text{ m s}^{-1}$.

0 3 . 1

Calculate V_1 .

[2 marks]

$V_1 =$ _____ V



Figure 4 shows the electron beam entering a uniform electric field between two horizontal plates.

At **B**, the electrons are moving horizontally.

The uniform electric field causes the electrons to accelerate.

0 3 . 2

Describe this acceleration.
Explain your answer.

[3 marks]

Question 3 continues on the next page

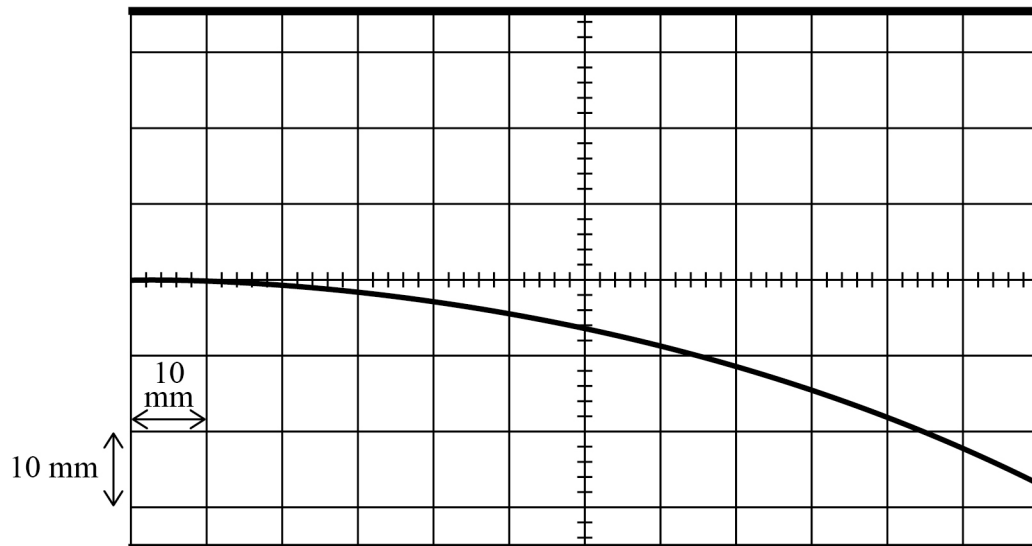
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The curved line on **Figure 5** shows the path of electrons on the fluorescent screen between the horizontal plates.

Figure 5

to scale



Each square on the screen measures 10 mm by 10 mm.

0 3 . 3 The horizontal component of an electron's velocity is $1.48 \times 10^7 \text{ m s}^{-1}$.

Show that the time taken for the electron to cross the screen is about 8 ns.

[1 mark]

0 3 . 4

The potential difference across the plates is equal to V_2 .

Determine V_2 .

[4 marks]

$V_2 =$ _____ V

0 3 . 5

The apparatus is enclosed in a glass tube.
Initially there is no air in the glass tube.
A fault occurs and some air leaks into the tube.

Explain **two** effects on the curved line seen on the screen.

[2 marks]

1 _____

2 _____

12

Turn over ►



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ANSWER IN THE SPACES PROVIDED**

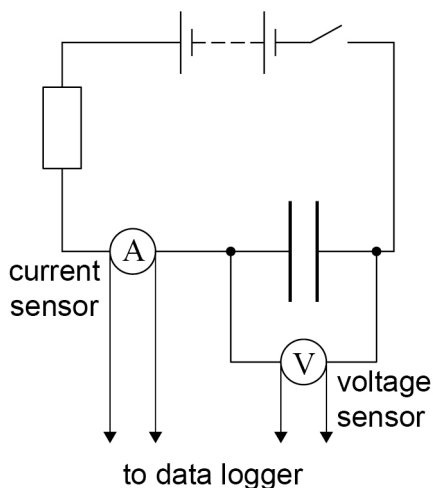


0 4

Figure 6 shows an arrangement for collecting data from a circuit that is used to charge a capacitor.

A data logger is connected to a current sensor and to a voltage sensor. The data are analysed.

Figure 6



The battery has an emf ε and negligible internal resistance. The resistance of the resistor is R and the capacitance of the capacitor is C .

The capacitor is initially uncharged.

The switch is then closed and the voltage V_c across the capacitor and the current I are recorded. Further readings of V_c and I are recorded at regular time intervals.

It can be shown that:

$$V_c = \varepsilon - IR$$

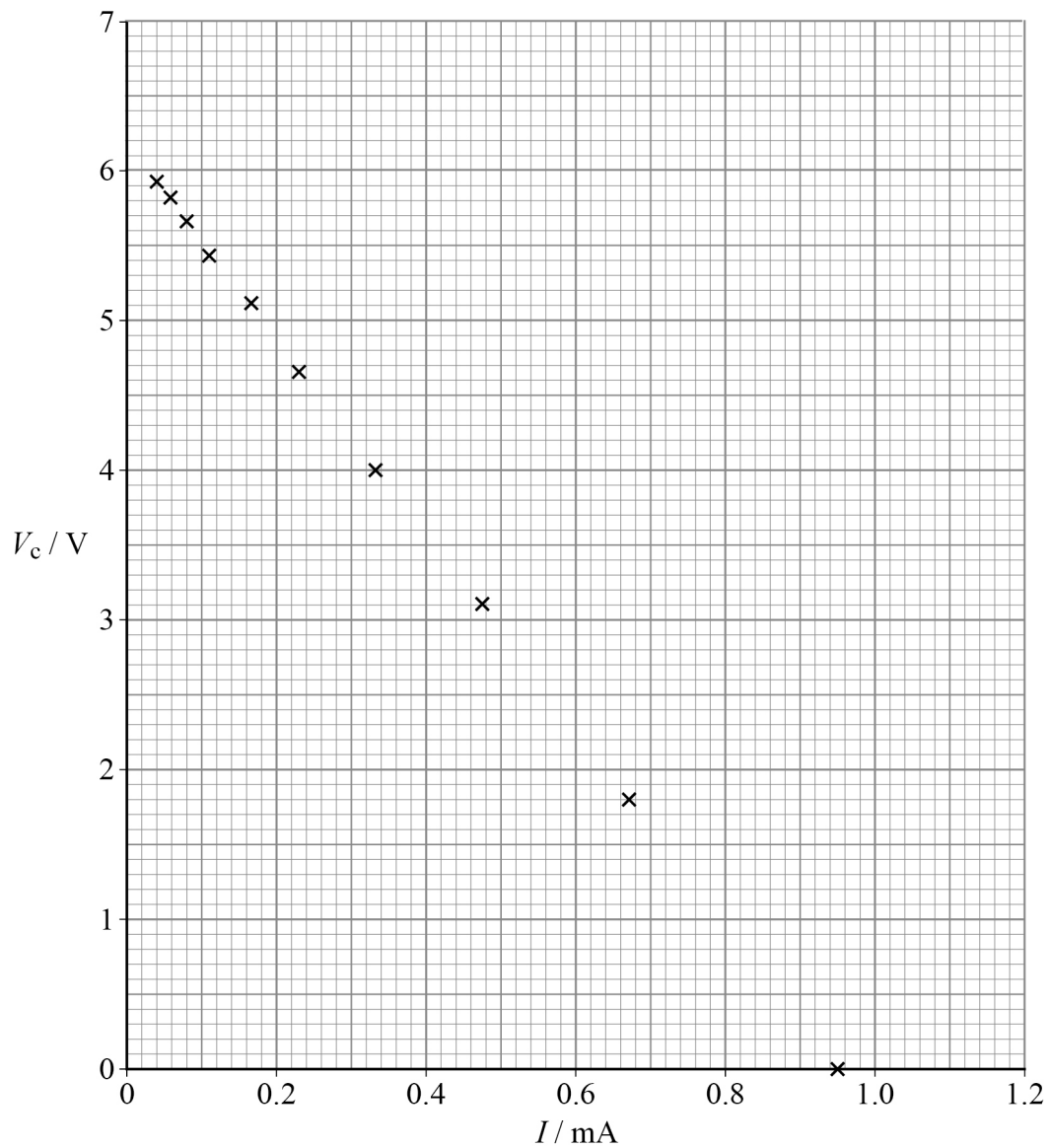
Question 4 continues on the next page

Turn over ►



Figure 7 shows the plot of V_c against I .

Figure 7



0 4 . 1 Determine $\mathcal{E}$.

[1 mark]

$\mathcal{E} =$ _____ V

0 4 . 2 Determine R .

[2 marks]

$R =$ _____ Ω



The data shown on **Figure 7** were recorded at time intervals of 5.0 ms.

0 4 . 3

Explain why the plotted points on **Figure 7** become closer together as V_c increases.

[1 mark]

0 4 . 4

Determine the time constant for the circuit in **Figure 6** on page 13.
Go on to calculate C .

[5 marks]

time constant = _____ s

C = _____ F

Question 4 continues on the next page

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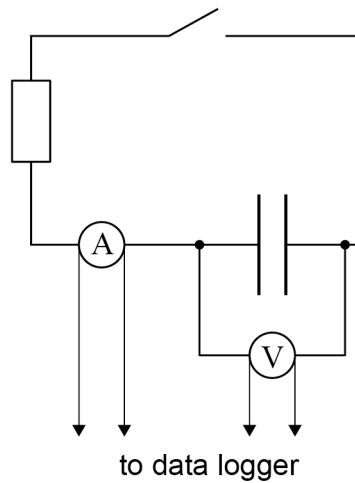


0 4 . 5

When the capacitor is fully charged the switch is opened. The battery is then removed from the circuit.

Figure 8 shows the new circuit for the charged capacitor.

Figure 8

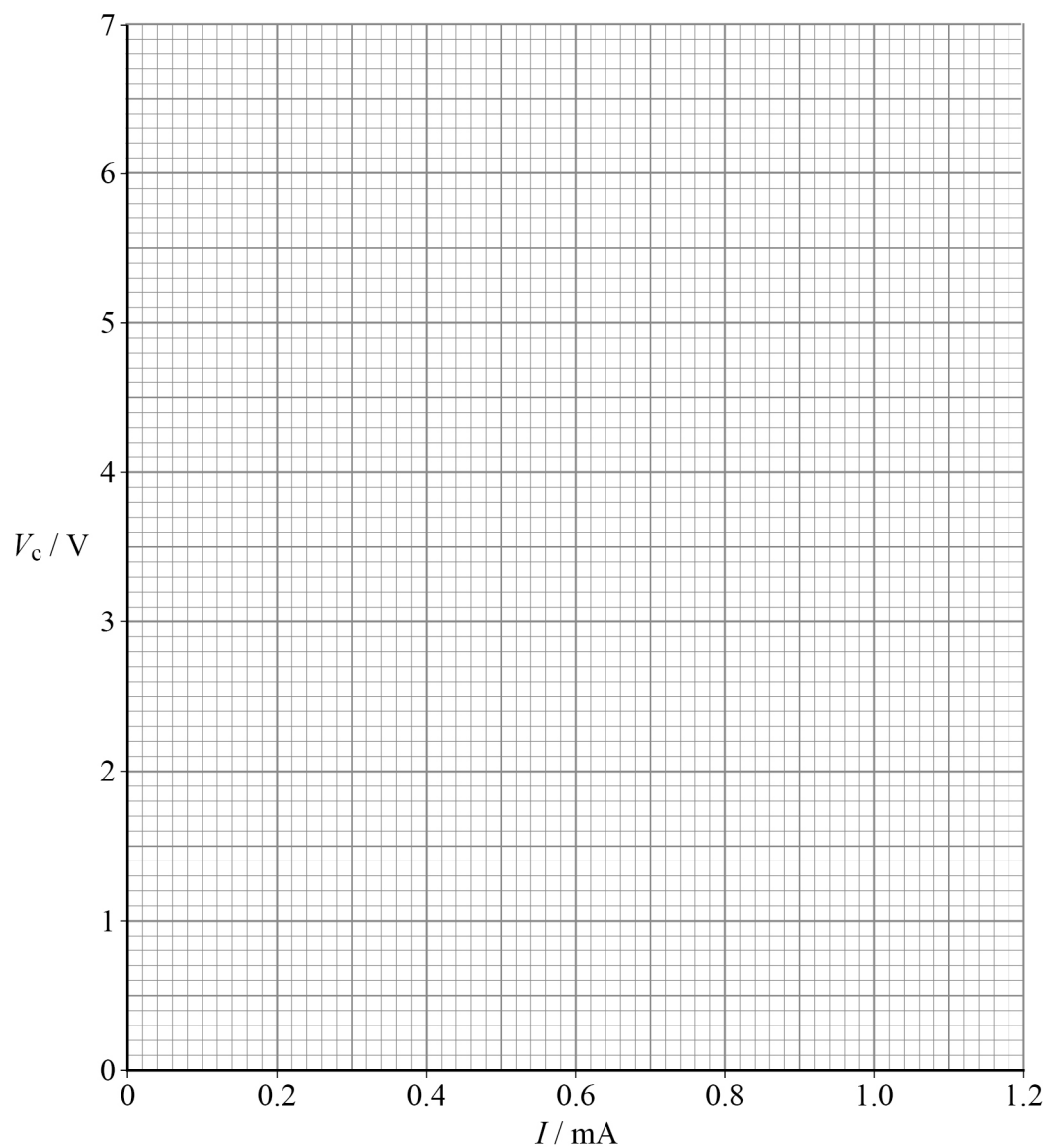


The switch is then closed and the capacitor discharges through the resistor.

Draw, on **Figure 9**, a line showing the variation of the magnitude of V_c with the magnitude of I for this experiment.

[2 marks]

Figure 9



0 4 . 6

Suggest why a data logger and sensors were used in these experiments rather than an ammeter, a voltmeter and a stopwatch.

[1 mark]

12

Turn over ►



0 5

Cobalt-60 (^{60}Co) is a radioactive isotope used in some school laboratories.

A school has a sealed ^{60}Co source with an activity of 62 kBq.

$$\begin{aligned}\text{half-life of } ^{60}\text{Co} &= 1.66 \times 10^8 \text{ s} \\ \text{molar mass of } ^{60}\text{Co} &= 60.0 \text{ g mol}^{-1}\end{aligned}$$

0 5 . 1

Calculate, in g, the mass of ^{60}Co in the sealed source.

[3 marks]

mass = _____ g

0 5 . 2

The school plans to replace the source when the activity of the ^{60}Co has decreased to 9.0 kBq.

Calculate, in years, the time before the source needs to be replaced.

[3 marks]

time = _____ years

6

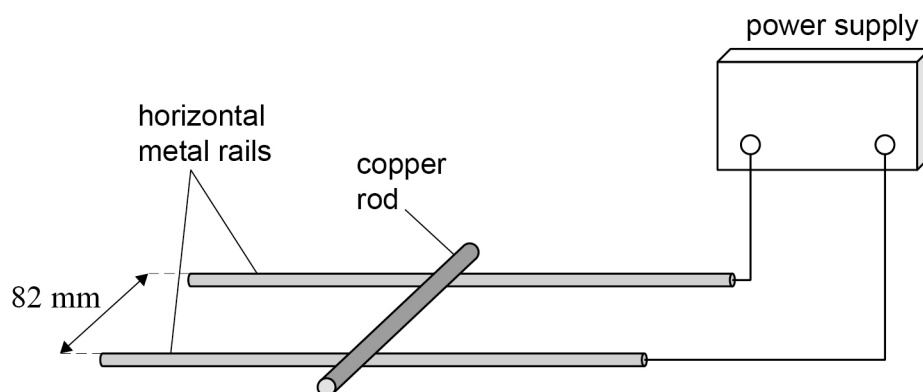
0 6 . 1 Define the tesla.

[1 mark]

For all parts of **Question 6**, assume that the Earth's magnetic flux density is negligible.

0 6 . 2 **Figure 10** shows a dc power supply connected to two horizontal metal rails. A copper rod rests on the rails, completing the circuit. Initially the power supply is off.

Figure 10



A uniform magnetic field is applied vertically downwards between the two rails. When the power supply is switched on, the copper rod rolls to the left.

Label with a + sign the positive terminal of the power supply in **Figure 10**. Explain your answer.

[2 marks]

Question 6 continues on the next page

Turn over ►



0 6 . 3

The power supply is switched off.

The rails are tilted at an angle of 10.0° to the horizontal.

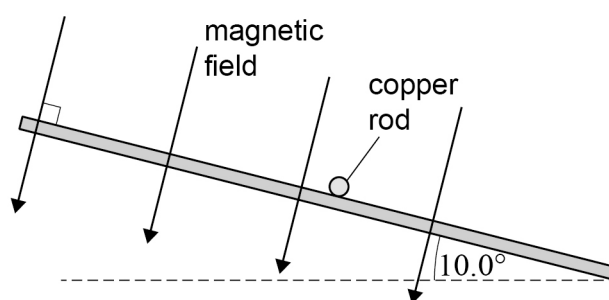
The direction of the magnetic field is changed so that it remains at 90° to the rails.

The power supply is switched on and the rod is placed on the rails. The combination of magnetic and gravitational forces acting on the rod keep it stationary.

Figure 11 shows a side view of the arrangement.

Figure 11

side view



The current in the rod is 2.75 A and the magnetic flux density is 0.11 T .

The distance between the rails is 82 mm .

Calculate the mass of the rod.

[3 marks]

mass = _____ kg



0	6	.	4
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The connections to the power supply are removed.

The rails remain tilted at an angle of 10.0° to the horizontal and the magnetic field is unchanged.

The rod is placed on the rails and released.

The rod accelerates down the rails.

State and explain what happens to the emf induced across the rod.

[2 marks]

8

Turn over for the next question

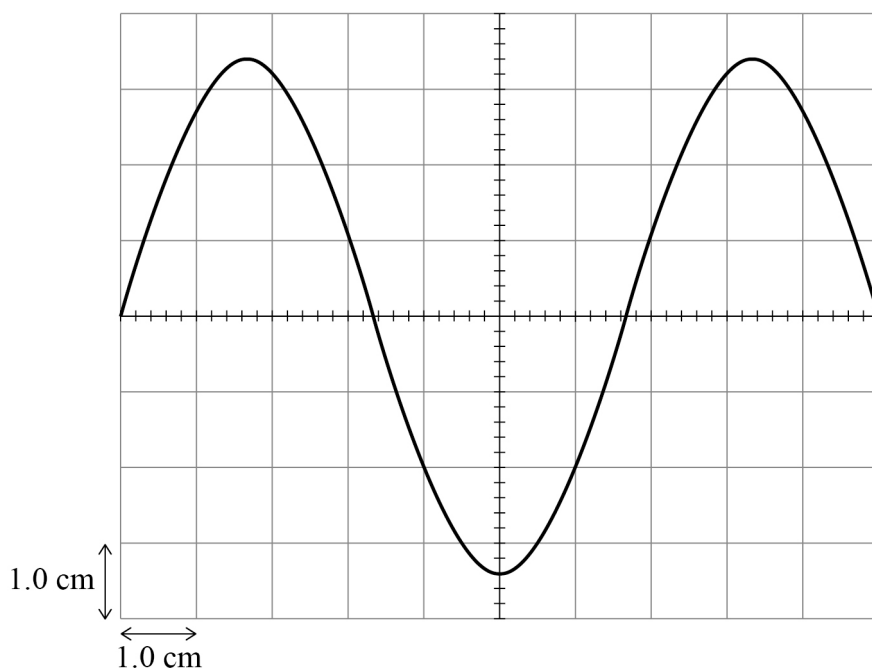
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0 7

Figure 12 shows the oscilloscope trace of the output of an ac power supply.

Figure 12



The root mean square (rms) voltage is 12.0 V and the frequency is 60.0 Hz.

0 7 . 1

Determine:

- the y -gain setting in V cm^{-1}
- the time-base setting in ms cm^{-1} .

[4 marks]

y -gain = _____ V cm^{-1}

time-base = _____ ms cm^{-1}



0 7 . 2

The output of the power supply is connected to the primary coil of a transformer.
The rms voltage output of the transformer is 3.0 V.
The number of turns on the secondary coil of the transformer is 4500

Calculate the number of turns on the primary coil.

[1 mark]

number of turns = _____

0 7 . 3

A resistor **R** is connected across the secondary coil.
The rms voltage across **R** is measured.

State the other measurements needed to determine the efficiency of the transformer.

[1 mark]

0 7 . 4

State **two** causes of inefficiency in a transformer.

[2 marks]

1 _____

2 _____

8

END OF SECTION A

Turn over ►



Section B

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown. 

If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown. 

You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

0 8

A bob on the end of a light string is rotated in a vertical circle at a constant angular speed of 5.5 rad s^{-1} .

The bob has a mass of 0.20 kg and the circle has a radius of 0.75 m .

What is the **minimum** tension in the string?

[1 mark]

A 2.0 N

☐

B 2.6 N

☐

C 6.1 N

☐

D 6.5 N

☐


0 9

A car is travelling on a circular track.

The car completes four revolutions of the track in one minute.

What is the average angular speed of the car?

[1 mark]**A** 0.030 rad s^{-1} ☐**B** 0.10 rad s^{-1} ☐**C** 0.42 rad s^{-1} ☐**D** 1.6 rad s^{-1} ☐**1 0**A charged particle is moving with a velocity v at right angles to a magnetic field of flux density B .The acceleration of the particle due to the magnetic field is a .

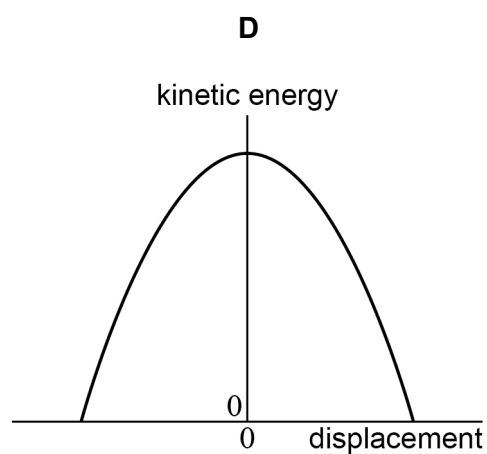
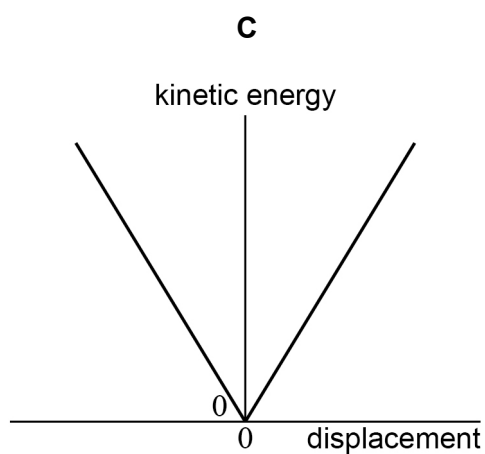
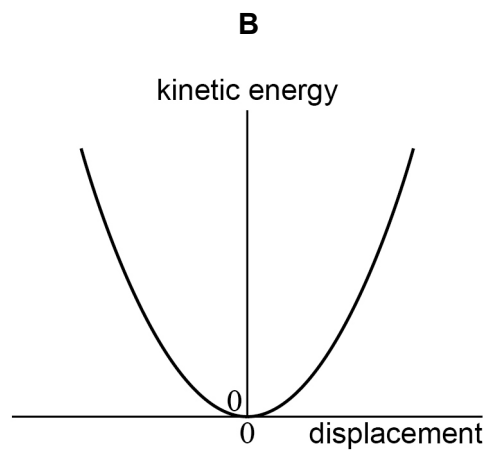
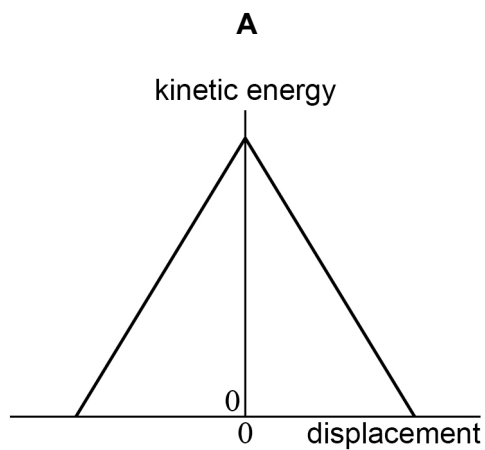
What is the specific charge of the particle?

[1 mark]**A** $\frac{Bv^2}{a}$ ☐**B** $\frac{Bv}{a}$ ☐**C** $\frac{a}{Bv}$ ☐**D** $\frac{a}{Bv^2}$ ☐**Turn over ►**

1 1

A mass on a spring undergoes simple harmonic motion.

Which graph shows the variation with displacement of the kinetic energy of the oscillating system?

[1 mark]**A** ☐**B** ☐**C** ☐**D** ☐

1 2

A satellite is in a circular orbit of radius R around a planet of mass M .
The orbital radius of the satellite is increased by a small distance Δr ,
where $\Delta r \ll R$.

The difference between the gravitational potentials of the two orbits is ΔV .

What is an approximation for ΔV ?

[1 mark]

A $\frac{GM\Delta r}{R^2}$ ☐

B $\frac{GM\Delta r}{R}$ ☐

C $\frac{GM}{R + \Delta r}$ ☐

D $\frac{GM}{\Delta r}$ ☐

1 3

A capacitor consists of two parallel plates with a dielectric between the plates.
The capacitor is connected to a battery and becomes fully charged.
The dielectric is then removed from between the plates without disconnecting the battery.

What happens to the charge on the plates and the energy stored on the capacitor?

[1 mark]

	Charge	Energy stored	
A	decreases	increases	☐
B	increases	decreases	☐
C	increases	increases	☐
D	decreases	decreases	☐



1 4

A charge of 1.5 mC is stored on a capacitor of capacitance 22 mF.
The capacitor is discharged until the pd across the capacitor is halved.

What is the change in the energy stored by the capacitor?

[1 mark]

A 13 μJ ☐

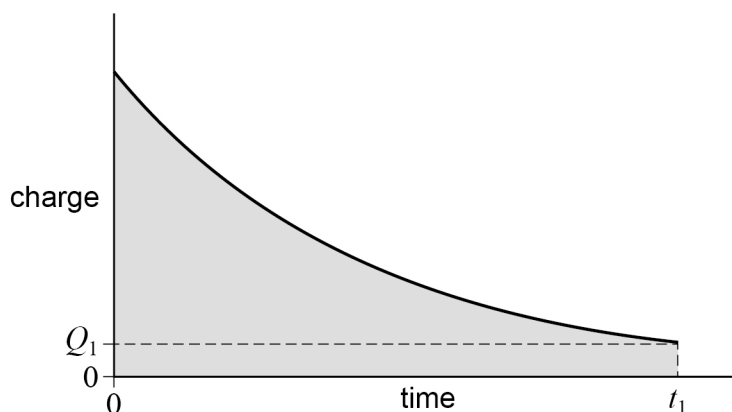
B 26 μJ ☐

C 38 μJ ☐

D 78 μJ ☐

1 5

A charged capacitor is discharged through a resistor of resistance R .
A graph is plotted of the charge on the capacitor against time.



After time t_1 the charge on the capacitor is Q_1 .

What is the capacitance of the capacitor?

[1 mark]

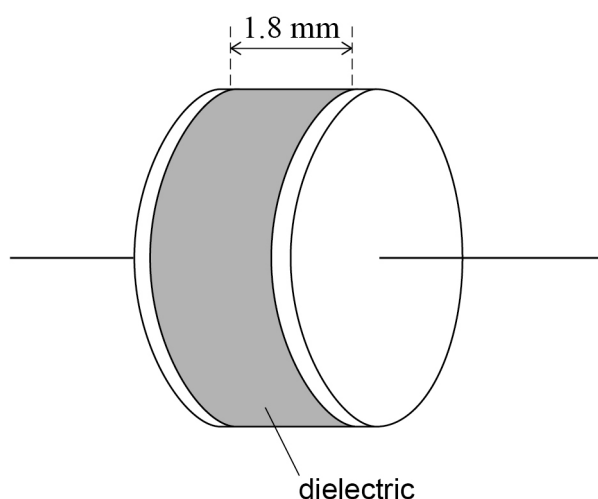
A $\frac{t_1}{R}$ ☐

B $\frac{\text{area under the graph}}{RQ_1}$ ☐

C $-\frac{\text{area under the graph}}{R \times \text{gradient at } (t_1, Q_1) \times t_1}$ ☐

D $-\frac{Q_1}{R \times \text{gradient at } (t_1, Q_1)}$ ☐

- 1 6** A parallel-plate capacitor is made from two identical circular discs that completely overlap.



A sheet of dielectric material of thickness 1.8 mm fills the space between the plates.
The material has a relative permittivity of 2.4
The capacitance of the capacitor is 2.2 nF.

What is the diameter of the discs?

[1 mark]

- A** 0.24 m ☐
- B** 0.49 m ☐
- C** 0.58 m ☐
- D** 1.17 m ☐

Turn over for the next question

Turn over ►



1 7

At a distance y from an isolated point charge, the electric potential is V and the electric field strength is E .

What are the electric potential and the electric field strength at a distance $2y$ from the charge?

[1 mark]

	Electric potential	Electric field strength	
A	$\frac{V}{2}$	$\frac{E}{2}$	☐
B	$\frac{V}{2}$	$\frac{E}{4}$	☐
C	$\frac{V}{4}$	$\frac{E}{2}$	☐
D	$\frac{V}{4}$	$\frac{E}{4}$	☐

1 8

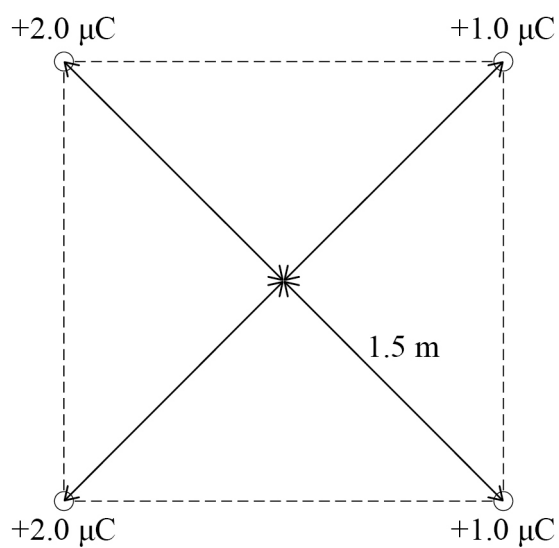
Which row identifies a vector quantity and a scalar quantity?

[1 mark]

	Vector	Scalar	
A	gravitational field strength	electric potential	☐
B	gravitational field strength	electric field strength	☐
C	gravitational potential	electric potential	☐
D	electric potential	gravitational field strength	☐



- 1 9** Four point charges are at the corners of a square.



What is the electric potential at the centre of the square?

[1 mark]

- A** $1.2 \times 10^4 \text{ V}$ ☐
- B** $2.4 \times 10^4 \text{ V}$ ☐
- C** $3.6 \times 10^4 \text{ V}$ ☐
- D** $4.0 \times 10^4 \text{ V}$ ☐

Turn over for the next question

Turn over ►



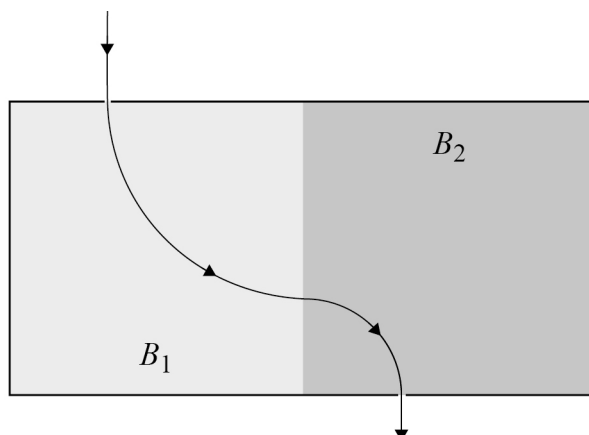
2 0

A proton passes through two evacuated chambers.

The chambers contain uniform magnetic fields of magnetic flux density B_1 and B_2 .

The path of the proton is at right angles to the magnetic field in each chamber.

The diagram shows the path of the proton from above.



$$x = \frac{\text{speed in second chamber}}{\text{speed in first chamber}}$$

Which row describes x and the directions of B_1 and B_2 ?

[1 mark]

	x	Direction of B_1	Direction of B_2	
A	>1	out of the page	into the page	☐
B	1	into the page	out of the page	☐
C	1	out of the page	into the page	☐
D	>1	into the page	out of the page	☐



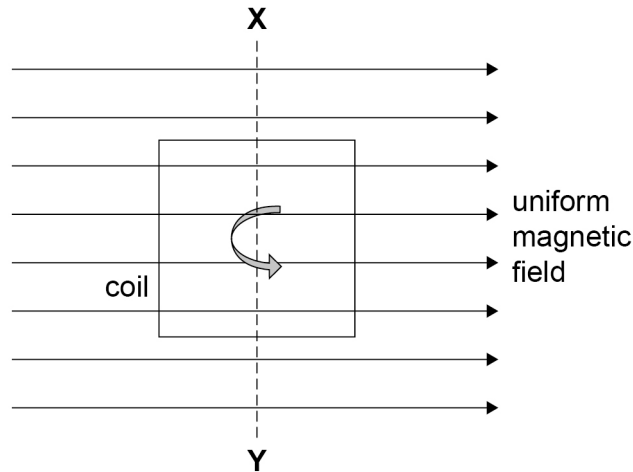
The following information is for Questions **21** and **22**.

A square coil of side 0.12 m has 20 turns.

The coil rotates with a frequency of 75 Hz in a uniform magnetic field.

The magnetic flux density is 45 mT.

The axis of rotation of the coil **XY** is at right angles to the magnetic field.



2 1 What is the maximum emf induced in the coil?

[1 mark]

- A** 0.30 V ☐
- B** 1.0 V ☐
- C** 6.1 V ☐
- D** 51 V ☐

2 2 What is the flux linkage at the instant when the minimum emf is induced in the coil?

[1 mark]

- A** 0 ☐
- B** 0.65 mWb ☐
- C** 13 mWb ☐
- D** 110 mWb ☐

END OF QUESTIONS



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